# Sheet 2
# Number System Operations

1. **Add the following Binaries**
   a. 11 + 1
   b. 11 + 11
   c. 101 + 101
   d. 10001 + 11101
   e. 10111 + 110001
   f. 1001.011 + 0100.110

2. **Subtract the following Binaries**
   a. 11 – 01
   b. 101 – 011
   c. 11011 – 10110
   d. 1110 – 0011
   e. 1100 – 0111

3. **Multiply the following Binaries**
   a. 11 * 11
   b. 111 * 101
   c. 1100 * 1101

4. **Divide the following Binaries**
   a. 110 / 11
   b. 10101 / 11
   c. 1100 / 100
   d. 11001 / 101
   e. 11000 / 1000

## 5. Add the following Octas

a. $(5)_8 + (4)_8$
b. $(243)_8 + (212)_8$
c. $(567)_8 + (243)_8$
d. $(1776)_8 + (345)_8$
e. $(147)_8 + (261)_8$
f. $(366.23)_8 + (243.62)_8$
g. $(53.012)_8 + (742.2)_8$

## 6. Subtract the following Octas

a. $(356)_8 - (157)_8$
b. $(2401)_8 - (1077)_8$
c. $(752)_8 - (361)_8$
d. $(1304007)_8 - (35006)_8$
e. $(272.423)_8 - (34.4444)_8$
f. $(543.125)_8 - (271.34)_8$

## 7. Multiply the following Octas

a. $(23)_8 * (6)_8$
b. $(15)_8 * (44)_8$
c. $(45.235)_8 * (7)_8$

## 8. Add the following Hexadecimals

a. $(3F8)_{16} + (3BF)_{16}$
b. $(8AB)_{16} + (B78)_{16}$
c. $(DAD)_{16} + (EB5)_{16}$
d. $(98F.A2)_{16} + (B11.94)_{16}$

## 9. Subtract the following Hexadecimals

a. $(9654)_{16} - (5321)_{16}$
b. $(974B)_{16} - (587C)_{16}$
c. $(FAB)_{16} - (AEF)_{16}$
d. $(F3A.251)_{16} - (FE.64)_{16}$

## 10. Multiply the following Hexadecimals

a. $(3A)_{16} * (2)_{16}$
b. $(3A)_{16} * (23)_{16}$
c. $(1E2)_{16} * (32)_{16}$
d. $(FE3)_{16} * (12D)_{16}$